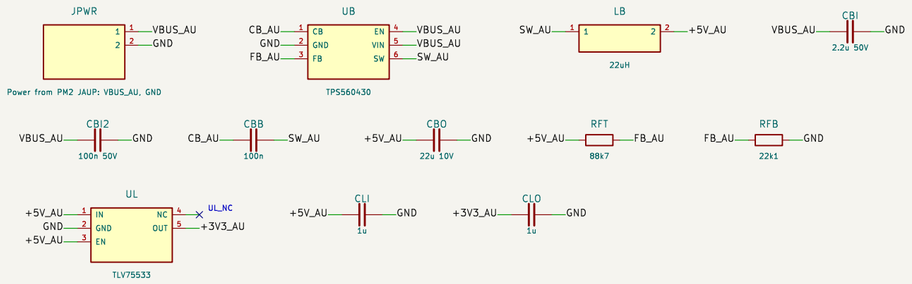
PF233-AU1: AU1 audio board: descriptor link, MSPM0G3507 voice synth, TPA2005D1 speaker amp, TPA6132A2 headphone amp, 3-position output selector, BMI270 yaw gyro

How to read: parts sit in functional blocks (dashed frames). Every pin ends in a short wire with its NET NAME; equal names are connected (also between blocks). A cross = intentionally unconnected pin.
Diode symbols point anode to cathode. Pin names are those of the library; items marked VERIFY in docs/ASSUMPTIONS.md are unverified. MODELED design: not built, not bench-tested.

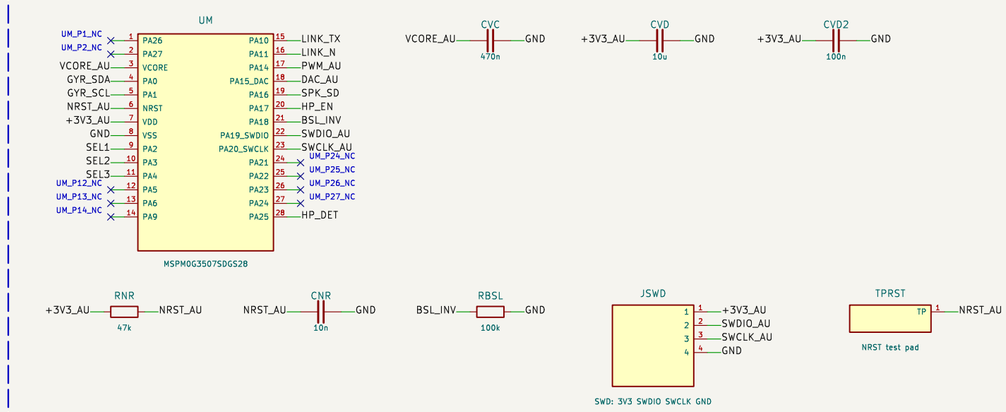
1 Power: pack -> 5 V buck -> 3.3 V LDO

JPWR from PM2-B JAUP (switched, fused 0.5 A). UB TPS560430 (RFT / RFB set 5.0 V, LB 22 uH), UL TLV75533 3.3 V.



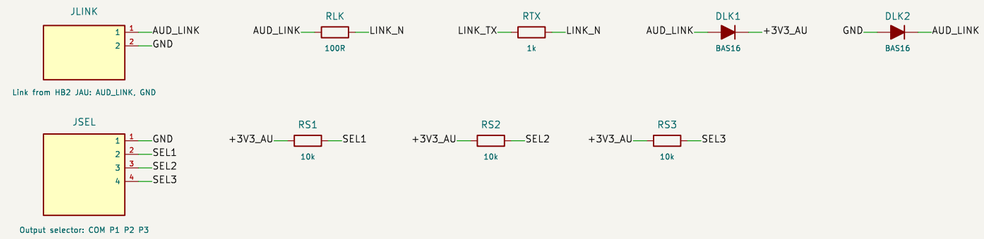
2 Controller

UM MSPM0G3507: DAC (PA15) for the voice, PWM (PA14) for the right channel, UART0 for the link; JSWD SWD pads, TPRST reset pad, RBSL keeps PA18 (BSL invoke) low.



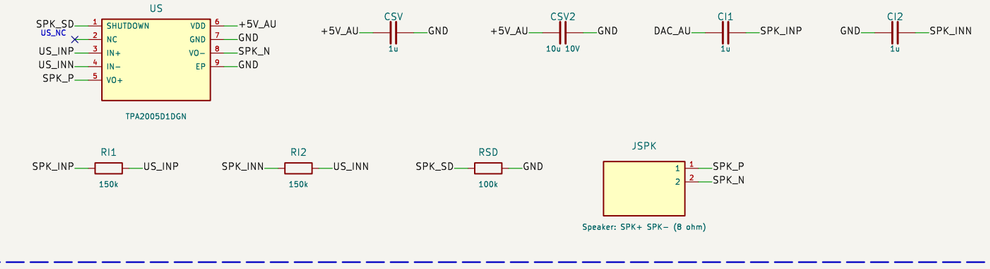
3 Descriptor link and output selector

JLINK from HB2-B JAU: half-duplex, TX through RTX 1 k, entry RLK 100 R with BAS16 clamps.
JSEL: 3-position selector to GND, pull-ups RS1-RS3 (AUDIO_ONLY / LED_ONLY / COMBINED).



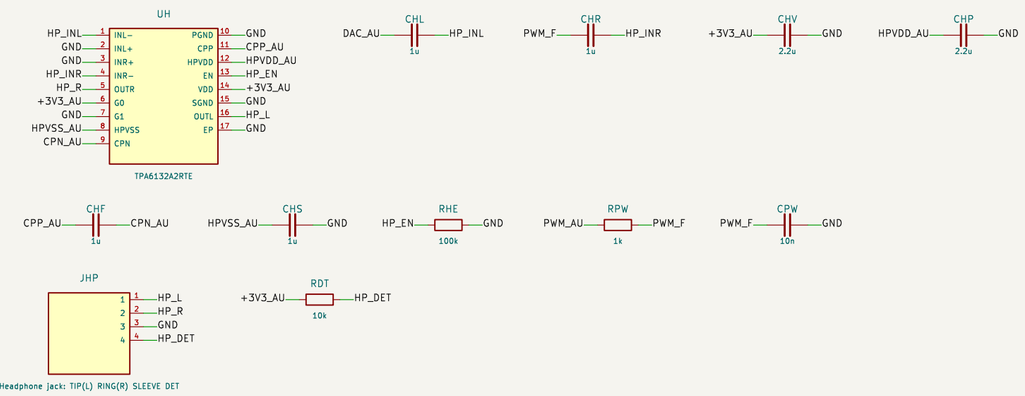
4 Speaker amplifier

US TPA2005D1, gain 2 x 150 k / RI = 2 V/V, input caps CI1 / CI2, SHUTDOWN pulled low (RSD) until the MCU enables it; JSPK 8 ohm speaker.



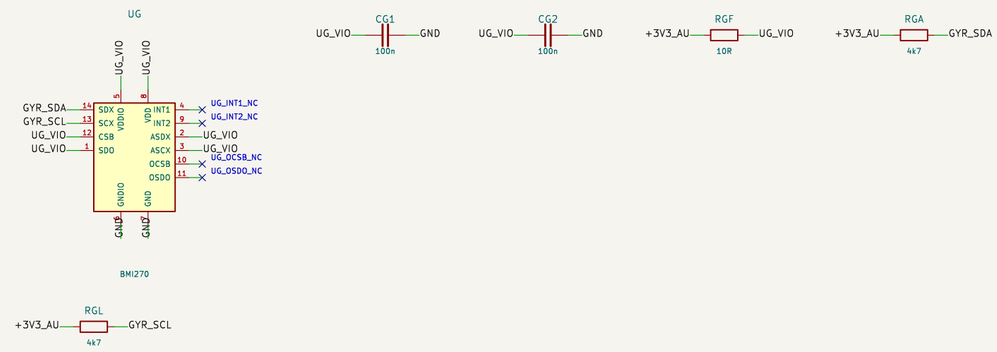
5 Headphone amplifier

UH TPA6132A2 DirectPath, 0 dB (G0 high, G1 low), single-ended inputs; left = DAC, right = RC-filtered PWM (RPW / CPW).
JHP: panel jack tip / ring / sleeve and plug-detect (RDT pull-up).



6 Yaw gyro (sweep direction, moved from CB1D)

UG BMI270 (alternate: ST LSM6DSO, same land) on I2C0, UM pins 4 / 5 (PA0 SDA / PA1 SCL); alternate address (SDO high); UG_VIO = VDD, VDDIO, CSB, SDO, pins 2 / 3, fed from +3V3_AU by RGF 10 R with CG1 / CG2 100 nF; RGA / RGL 4k7 pull-ups. The yaw rate goes to the head in the link reply.



Generated: src/pcb/sch_func.py; checked against the netlist.
MODELED design: not built, not bench-tested.
AU1 audio board: descriptor link, MSPM0G3507 voice synth, TPA2005D1 speaker amp, TPA6132A2 head
ARX / PHI-FOLD 233

Sheet: /
File: PF233-AU1_functional.kicad_sch

Title: PF233-AU1 functional schematic

Size: User	Date: 2026-10-26	**Rev: r3**
KiCad E.D.A. 9.0.9		Id: 1/1